LOGIC

PROPOSITIONS AND COMPOUND STATEMENTS

- A proposition/statement is a declarative sentense that is either true or False but not both.

E.g

1.	Nairobi is the capital of Kenya	True, T
2	1+1 = 2	True T
3	8 < 5	False, F
4	2+2=3	False, F

Some sentences are not propositions eg

1. What is your name ?.
2 what time is it ?.
3. x+4 = 2
4 x+y = z
5. Read this carefully.

Sentence 1, 2 & 5 are not propositions because they are not declarative sentences. Sentences 3 & 4 are not propositions because they are neither true or False since the variable in these sentences have not been assigned values.

Letters are used to denote propositions just as they are used to denote variables. The conventional / convinient letters used for this purpose are P, q, r, s, ...

IF a proposition is true, the truth value is true denoted by T and if it is False, its true value is False denoted by F

The area of Logic that deals with propositions is called propositional Calculus or propositional Logic

Compound propositions - are those that are composed of subpropositions and seriors Connectives / logical ~~operation~~ operator
1. Roses are red and violets are blue.
2 John is intelligent or studies every night

NB// The truth value of a compound proposition is completely determined by truth values of its sub-propositions together with the way they are connected

Basic logical operations and their connectives

	Operation	Logical operator / connectives	operator symbol
1	Negation	Not	¬ or ~
2	Conjuction	And	∧
3	Disjunction	or	∨

¬p or ~p reads as "Not p"

1. Negation
Given any proposition P, the negation of P denoted by ¬p or ~p can be formed by writing
"It is not the case that p"
"It is false that p" or if possible it can be written by inserting in p the word "not"

EXAMPLE

Find the negation of the proposition

1 Nairobi is in Kenya.

Soln

It is not the case that Nairobi is in Kenya
It is False that Nairobi is in Kenya
Nairobi is not in Kenya.

2 "Today is Friday"

Soln

It is not the case that today is Friday
It is False that today is Friday
Today is not Friday.

NB// The truth value of the negation of P is alway opposite that of P (see table below)

P	$\neg P$
T	F
F	T

This is called the truth table for negation of a proposition

- A truth table displays the relationships between the truth values of propositions
- Truth tables are especially valuable in the determination of the truth values of propositions constructed from simple propositions.

2 CONJunction

If p and q are propositions, the conjuction of p and q denoted by $p \wedge q$ is the proposition "p and q"

- The conjuction $p \wedge q$ is true when and only when both p and q are true and False otherwise

Truth table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F

3 DisJunction

If p and q are propositions, the disJunction of p and q denoted by $p \vee q$ is the proposition of "p or q"

- The disJunction $p \vee q$ is False when and only when both p and q are False otherwise it is true.

Truth table for the DisJunction of two propositions

P	P	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

EXAMPLE

Here we mean that students who have taken both Calculus and Computer science can take the class, as well as the students who have taken only one of the two subjects

On the other hand, we are using the exclusive or when we say "Students who have taken Calculus or Computer science course ~~but not both~~ cannot take the class.

For the class here we mean that students who have taken both Calculus and computer science course cannot take the class.

On those who have taken exactly one of the two courses can take the class.

- Similarly in a Menu at a restaurant states "soup or salad comes with an entree" The restaurant means that Customers can have either soup or salad but not

Hence this is an exclusive rather than inclusive

NB// p but q means p and q

Neither p or q means ~p and ~q

- Let p and q be propositions. The exclusive or of p and q denoted by $p \oplus q$, is the proposition that is true When exactly one of p and q is true and is False otherwise

Truth table for the exclusive or of two proposition

P	q	$P \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Assignment

1 $(p \wedge q) \vee r$

2 $(\sim q \vee r) \wedge r$

3 $\sim q \wedge r$

where: r – all sides of equilateral Δ are equal

p – Month of august ends 31

q – 2 < – 8

CONSTRUCTION OF TRUTH TABLES FOR COMPOUND PROPOSITIONS

* Start with the variables in the first column
* If there are n variables, 2^n rows are required to allow for all possible combinations of T and F
* Truth value of proposition appears in the last column

EXAMPLE

Construct a truth table for the proposition $\neg(p \wedge \neg q)$

Soln

P	q	¬q	P∧¬q	~(P∧¬q)
T	T	F	F	T
T	F	T	T	F
F	T	F	F	T
F	F	T	F	T

Assignment

Construct a truth table for the proposition

i) ¬(¬P∧¬q)

ii) ¬(¬P∨¬q)

Assignment 2

1. Construct the truth table for each of the following compound propositions

a) (P∨q)∨r

b) (P∧q)∨r

c) (P∧q)∨¬r

d) (P∨q)∧r

e) (P∧q)∧r

f) (P∧q)∨¬r

Tautologies and contradictions

A tautology is a compound proposition that is ~~and~~ true for any truth value of its variable

fallacy Contradiction – is a compound proposition that is false

Contingency – is a compound proposition that is neither a tautology nor a contradictions.

Example:

1) "P or not P" i.e $P \vee \sim P$ is a tautology

P	$\neg P$	$P \vee \sim P$
T	F	T
F	T	T

2. "P and not P" i.e $P \wedge \sim P$ is a contradiction

P	$\sim P$	$P \wedge \sim P$
T	F	F
F	T	F

NB// The negation of an tautology is a contradiction the negation of a contradiction is a tautology.

LOGICAL EQUIVALENCE

Two proposition P and Q are said to be logically equivalent denoted by $P \equiv Q$ or $P \Leftrightarrow Q$ if and only if the column giving their truth values agree or have identical values.

Examples

$$\sim(P \wedge Q) \equiv \sim P \vee \sim Q$$
$$\sim(P \vee Q) \equiv \sim P \wedge \sim Q$$

* De-Morgan's law's

a) Show that $\sim(p \vee q) \equiv \sim p \wedge \sim q$

soln

p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

NB// Note that in each row, the truth value of $\sim(p \vee q)$ is the same as the truth value of $\sim p \wedge \sim q$. Hence $\sim(p \vee q)$ and $\sim p \wedge \sim q$ are logically equivalent

exercise

(b) Show that $\sim(p \wedge q) \equiv \sim p \vee \sim q$

soln

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

$\therefore \sim(p \wedge q) \equiv \sim p \vee \sim q$

Example — Distributive law of disjunction or conjunction

Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent

Soln

p	q	r	$q \wedge r$	$p \vee q$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T
T	F	F	F	T	T	T
T	F	F	F	T	T	T
T	F	T	F	T	T	T
F	T	F	F	T	F	F
F	T	T	T	T	T	T
F	F	T	F	F	F	F
F	F	F	F	F	F	F

same truth values for logically equivalent

EXERCISE

a) use truth tables to verify the associative laws
   i) $(p \vee q) \vee r \equiv p \vee (q \vee r)$
   ii) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
b) use the truth tables to verify the distributive Law.
   Pr $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

CONDITIONAL AND BI-CONDITIONAL STATEMENTS

Condition

A conditional statement is a statement of the form "If P then q". It is called an implication

Defination

If P and q are statements/propositions, the condition statement "If P then q" or P implies q is denoted by $P \rightarrow q$
It is False when p is true and q is false otherwise it is true.
We call P the hypothesis/antecedent/premise and q is called the Consequent/conclusion

Truth table to $P \rightarrow q$

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

You will encounter most if not all of the following ways to express this implication

"If P, then q" "P implies q"
If P, q" " P only if q
" P is sufficient for q" "a sufficient condition for q is P"
"~~If~~ "q if p" "q whenever p"
" q when p" "q is necessary for p"
"a necessary condition for p is q" "q follows from p"

E.g

1. "If I am elected, then I will lower taxes"
2. "For John to get a Job, it is sufficient for him to ____ "
3. "If you get 100% on the final exam, then you will get an A".

NB// Definition of Conditional statement in Mathematics is More general than the Meaning attached to same statement false in English

Example

1. If today is Friday, then $2+3=5$ is true from that definition of implication, since its conclusion is truth (Truth Value of the hypothesis does not Matter)
2. IF today is Friday, then $2+3=6$ is true everyday except Friday, even though $2+3=6$.

Exer

1. Use the truth table to show the logical equivalent of the statement forms, $p \vee q \to r$ and $(p \to r) \wedge (q \to r)$
   i.e
   show that
   $$(p \vee q) \to r \equiv (p \to r) \wedge (q \to r)$$
2. Show that $p \to q \equiv \sim p \vee q$

Negation of conditional statement

The negation of "If p then q" is logically equivalent to; "p and not q"

Example

Negate the following

1. If My car is in the garage, then I cannot get to class.

soln

My car is in the garage and I can get to class

2. If John lives in Nchiru, then she lives in Kenya.

John lives in Nchiru and she does not Live in kenya.

BI-CONDITIONAL STATEMENTS

Let p and q be propositions. The biconditional of p and q is "p if and only if" and is denoted as $p \iff q$ This proposition is true when p and q have the same truth Values and False otherwise.

Truth table for Biconditional $p \iff q$

p	q	$p \iff q$
T	T	T
T	F	F
F	T	F
F	F	T

- In essence, $p \iff q$ is the conduction of two conditions statements; $p \to q$ and $q \to p$ i.e

$p \Rightarrow q \wedge q \rightarrow p$

Other common ways to express $p \Leftrightarrow q$ are;

"p is necessary and sufficient for q"
"If p then q, and conversely"
"p if q"
"p iff q"

Example

Rewrite the following statement as a conjuction of two, if then statements "You can take the flight if and only if you buy a ticket."

Soln

If you can take the flight, then you buy a ticket; and if you buy a ticket you can take the flight.

Precedence of logical operations

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\Leftrightarrow$	5

Exercise

Use the truth tables to show that the following propositions are tautologies

a) $[\neg p \wedge (p \vee q)] \rightarrow q$

b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

c) $[p \wedge (p \rightarrow q)] \rightarrow q$

d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

e) $[\neg (p \rightarrow q) \iff [p \wedge (\neg q)]$

f) $[p \vee (q \vee r)] \iff [(p \vee q) \vee r]$

g) $[p \wedge (q \vee r) \iff [(p \wedge q) \vee (p \wedge r)]$

h) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

LOGICAL EQUIVALENCE

	Equivalence	Name
1.	$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity Laws
2	$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination Laws
3	$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent Law
4	$\neg(\neg p) \equiv p$	Double negation Law
5	$p \vee q = q \vee p$ $p \wedge q = q \wedge p$	commutative Laws

6 $(p \vee q) \vee r \equiv p \vee (q \vee r)$ Associative laws
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

7 $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ Distributive
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ Laws

8 $\neg(p \wedge q) \equiv \neg p \vee \neg q$ Demorgans'
$\neg(p \vee q) \equiv \neg p \wedge \neg q$ Laws

9 $p \vee (p \wedge q) \equiv p$ Absorption Laws
$p \wedge (p \vee q) \equiv p$

10 $p \vee \neg p \equiv T$ Negation laws
$p \wedge \neg p \equiv F$

Logical Equivalence Involving Implications

$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \vee q \equiv \neg(p \rightarrow q$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$

$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Logical equivalence Involving Bi-Conditionals

$p \Longleftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \Longleftrightarrow q \equiv \neg p \Longleftrightarrow \neg q$
$\neg(p \Longleftrightarrow q) \equiv p \Longleftrightarrow \neg q$

Examples

1. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

Soln a) use truth table

b) $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ and demogen Law

$\equiv$ ~~$\neg p \wedge (\neg$~~

$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$ 1st " "

$\equiv \neg p \wedge (p \vee \neg q)$ double negation

$\equiv (\neg p \wedge p) \vee [\neg p \wedge \neg q]$ Distribution Law

$\equiv F \vee (\neg p \wedge \neg q)$ since $\neg p \wedge p = p \wedge \neg p \equiv F$

$\equiv (\neg p \wedge \neg q) \vee F$ commutative law for disjunction

$\equiv \neg p \wedge \neg q$ from identity Law for F

Consequently $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

2 Show that $(p \wedge q) \rightarrow (q \vee q)$ is a tautology

a) use truth table

b) $(p \wedge q) \rightarrow (q \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$

$\equiv (\neg p \vee \neg q) \vee (p \vee q)$ Demorgan's Law

$\equiv (\neg p \vee p) \vee (\neg q \vee q)$ By associative & commutative law & disjunction

$\equiv T \vee T$

$= T$ By domination law,

Exercise

a) show that $\neg p \Longleftrightarrow q$ and $p \Longleftrightarrow \neg q$ are logically equivalent

b) show that $\neg(p \Longleftrightarrow q)$ and $\neg p \Longleftrightarrow q$ are logically equivalent

c) show that $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$
d) show that $\neg p \to (q \to r) \equiv q \to (p \vee r)$
e) show that $\neg (p \iff q) \equiv p \iff \neg q$

f) show that $(p \vee q) \wedge (\neg p \vee r) \to (q \vee r)$ is a tautology
g) show that $(p \to q) \wedge (q \to q) \to (p \to r)$ " "

CONVERSE, CONTRAPOSITIVE AND INVERSE

Staring with a Conditional statement $p \to q$
we can form the following special Conditional statements:

1 Converse of $p \to q$ is $q \to p$
2 Inverse of $p \to q$ is $\neg p \to \neg q$
3 Contrapositive of $p \to q$ is $\neg q \to \neg p$

NB//
(1) The contrapositive always has same truth values as the statement $p \to q$
2 Converse and inverse of a Conditional statement are equivalent But neither is equivalent to the original Conditional statement

EXAMPLE

write the Contrapositive, Converse, and the inverse of the Conditional statement
1) "The birds shelter whenever it is raining"
Same as: If it is raining birds shelter.
Contrapositive: If the birds do not shelter,

then it is not raining

Converse: If the birds shelter then it is raining

Inverse: If it is not raining then the birds do not shelter.

2 The home team wins whenever it is raining

This is same as: If it is raining, then home team wins

Contrapositive: If the home team does not win, then it is not raining

Converse: If the home team wins, then it is raining

Inverse: If it is not raining, then home team does not win

QUESTION

State the Converse, Contrapositive and inverse of the each of these implication

a) If it snows tonight, I will stay at home.

b) If it rains today, I will plant tomorrow

(c) Converse

(ii) Contrapositive

(iii) Inverse.

(c) I go to beach whenever it is a sunny summer day

(d) A positive integer is a prime only if it has no divisors other than 1 and itself

e) ~~I come to~~

e) whenever I stay up late, it is necessary that I sleep until noon.

Necessary and Sufficient Conditions

If P and Q are statements then

- P is a sufficient condition for Q Means "If P, then Q"
- P is necessary condition for Q Means "If not P, then not Q" or "If Q then P"
- Consequently P is a necessary and sufficient condition for Q Means "P If and only if Q"

Example

Re-write the following as "If then ..." statement.

1. John's birth on Kenyan soil is a sufficient condition for him to be a Kenyan citizen

Ans

If John's birth is on Kenyan soil, then he is Kenyan citizen

2. Mary's attaining 18 yrs is a necessary condition for her to have I.D.

Exam

Translating English sentences

Translating sentences in natural language e.g English into logical expression is essential especially when specifying both hardware and software system.

- Systems and software engineers take requirements in natural language and produce precise and unambiguous specifications that can be used as the basic for system development.

e.g

Translate the following english sentences into logic expressions

1) "You can access the internet from campus only if you are a computer Major or you are not a freshman.

Soln

Let a, b and c represent

a - "You can access the internet from campus"

b - "You are a computer science student/Major"

c - "You are a freshman"

Hence $a \rightarrow (b \vee \neg c)$

2 "You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."

Soln

Let q, r and s represent

q - "You can ride roller coaster"

r - "You are under 4 feet tall"

s - "You are older than 16 yrs old"

Hence $(r \wedge \neg s) \rightarrow q$

Exercise

1. Consider the following proposctions
p: I will eat My socks
q: It is raining
r: I will go shopping

Represent the following sentence using logical connectives and letters p, q or r

a) If it is raining, I will eat my socks but I will not go shopping.
b) It is either raining or I will go shopping but I will eat My socks if it is raining
c) That it is raining is necessary and sufficient condetions for me not to go shopping

2. let p and q be the propositions
p: It is below freezing
q: It is snowing

Write these propositions using p and q and logical connectives
a) It is not below freezing and snowing
b) It is below freezing but not snowing
c) It is not below freezing and it is not snowing
d) It is either snowing or below freezing
e) If it is below freezing, it is also snowing
f) If it is either below freezing or it is snowing, but it is not snowing if it is below freezing

3 Let p, q and r be the propositions
p: you have the flu
q: you miss the final examination
r: you ~~Miss~~ pass the course

Express each of these propositions as an English sentence

a) $p \rightarrow q$

b) $\neg q \leftrightarrow r$

c) $p \vee q \vee r$

d) $(p \wedge q) \vee (\neg q \wedge r)$

QUANTIFIERS

Some statements contain variables and hence they are true or false when appropriate values are substituted for the variables

E.g $\underbrace{15}_{\text{predicate}} \leq \underbrace{x}_{\text{subject}}$ Such are called predicates

There are two types of quantifiers.

1. Universal quantifiers ($\forall$)

The notation $\forall x\, P(x)$ denotes the universal quantifications of $P(x)$. Here $\forall$ is called the universal quantifiers. The proposition $\forall x\, P(x)$ is read as "for all x $(P(x))$" or "For every x $P(x)$"

This quantifiers asserts that a certain property hold for all objects of certain type

Example

1. Let $P(x)$ be statement "$x+1 > x$". What is the truth value of quantification $\forall x P(x)$, where the universe of discourse consists of all real no's.

Soln

Since $P(x)$ is true for all real numbers, x, the quantifications $\forall x P(x)$ is true.

2. Let $Q(x)$ be the statement "$x < 2$". What is the truth value of the quantification $\forall x Q(x)$, where the universe of discourse consists of all real no's.

Soln

$Q(x)$ is not true for every real no's x, Since for $Q(3)$ is false.
Thus $\forall x Q(x)$ is false.

3. What is the truth value of $\forall x P(x)$ where $P(x)$ is statement $x^2 < 10$ and the Converse of discourse consists of the positive integer not exceeding 4?

Soln

The statement $\forall x P(x)$ is the same as conjuction $P(1) \wedge P(2) \wedge P(3) \wedge P(4)$
Since the universe of discourse consists of integers $1, 2, 3, \& 4$.
Since $P(4)$, which is statement $4^2 < 10$, is false, it follows that $\forall x P(x)$ is False.

2. Existential Quantifier ($\exists$)

The existential quantification of $P(x)$ is the proposition "There exists an element x in the universe of discourse such that $P(x)$ is true."

We use the notation $\exists x\, P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called existential quantifier. $\exists x\, P(x)$ It is read as, "There is an x such that, "There is at least one x such that $P(x)$."

Example

1) Let $P(x)$ denote statement "$x > 3$" what is the truth value of quantification $\exists x\, P(x)$, where the universe of discourse consists of all real numbers.

Soln

Since $x > 3$, it is true for instance $x = 4$, the existing quantification of $P(x)$ which is $\exists x\, P(x)$ is true.

2) Let $Q(x)$ denotes statements $x = x + 1$. what is the truth value of the quantification $\exists x\, Q(x)$, where the universe of discourse consists of all real no.s.

Soln

Since $Q(x)$ is false $\forall$ real no. x the existential quantification of $Q(x)$, which is $\exists x\, Q(x)$ is false.

Statement	When true	When False
$\forall x P(x)$	$P(x)$ is true for every x	There is an x for which $P(x)$ is False for every x
$\exists x P(x)$	There is an x for which $P(x)$ is true	